

Chapter 1 – Introduction to Ubuntu Server

1.1 What is Ubuntu Server?

Ubuntu Server is a free, open-source Linux operating system developed by **Canonical Ltd.** It is designed specifically to run servers and network services instead of providing a desktop environment.

Unlike Ubuntu Desktop, Ubuntu Server installs with a **command-line interface (CLI)** by default, making it lightweight, stable, and suitable for enterprise environments.

Ubuntu Server is one of the most widely used Linux server operating systems because it is reliable, secure, and supported by a large community.

1.2 Why Learn Ubuntu Server?

Most enterprise services run on Linux servers.

Examples include:

- Web Servers
- Database Servers
- Mail Servers
- DNS Servers
- Cloud Platforms
- Docker Containers
- Kubernetes
- SIEM Platforms
- Wazuh
- Splunk
- Elasticsearch

If you want to become a SOC Analyst, Security Engineer, or System Administrator, learning Ubuntu Server is an essential skill.

1.3 Why Do We Use Ubuntu Server?

Ubuntu Server offers several advantages:

- Free and Open Source
- Lightweight
- Long-Term Support (LTS)
- Excellent Security
- Stable Performance
- Large Community Support
- Enterprise Ready
- Easy Package Management using APT

These features make Ubuntu Server suitable for both home labs and production environments.

1.4 Ubuntu Server vs Ubuntu Desktop

Ubuntu Server	Ubuntu Desktop
Command Line Interface	Graphical Desktop
Lower Resource Usage	Higher Resource Usage
Designed for Servers	Designed for Personal Computers
Better Performance	More User Friendly
Common in Data Centers	Common on Home PCs

1.5 Common Uses of Ubuntu Server

Ubuntu Server can host many services.

Examples include:

- Apache Web Server
- Nginx
- MySQL
- PostgreSQL
- Docker
- Kubernetes
- Active Directory Alternatives
- File Servers
- VPN Servers
- Wazuh
- Jenkins
- GitLab

One physical server can host multiple services depending on available hardware.

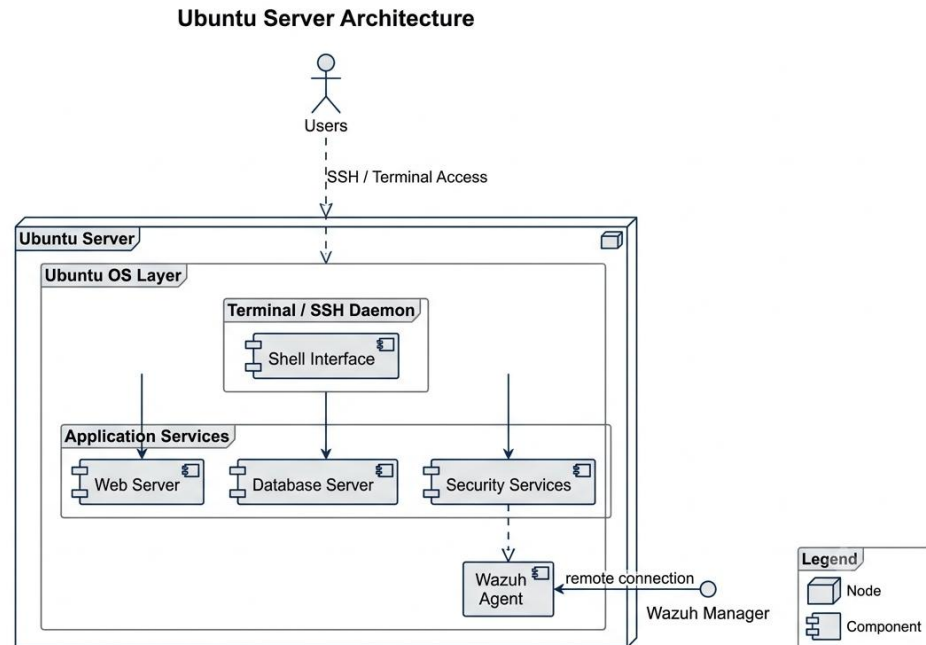
1.6 Why Ubuntu Server for Wazuh?

Wazuh officially recommends Ubuntu Server LTS versions because they provide:

- Stable package repositories
- Long-term security updates
- Better compatibility
- Production-grade reliability

Using a supported Ubuntu LTS release helps avoid compatibility issues during installation.

1.7 Ubuntu Server Architecture



1.8 Ubuntu Server Features

Ubuntu Server includes:

- Multi-user support
- Secure Shell (SSH)
- Package Manager (APT)
- Firewall (UFW)
- Networking tools
- Systemd service management
- File permissions
- User management
- Cron jobs
- Logging

These tools form the foundation of Linux system administration.

1.9 Real-World Applications

Ubuntu Server is used in:

- Banks
- Government organizations
- Universities
- Cloud providers
- Data centers
- SOC environments
- Cybersecurity labs
- Software development companies

Many cybersecurity tools are designed to run on Linux servers.

1.10 Key Terms

Term	Description
Server	A computer that provides services to other computers.
CLI	Command Line Interface used to interact with the operating system.
SSH	Secure Shell protocol for remote administration.
LTS	Long-Term Support release with extended security updates.
Package	Software installed using a package manager such as APT.
Service	A background process managed by systemd.

Chapter Summary

In this chapter, you learned:

- What Ubuntu Server is.
- How it differs from Ubuntu Desktop.
- Why it is widely used in enterprise environments.
- Why it is the preferred platform for Wazuh and many other server applications.
- Common services hosted on Ubuntu Server.

One-Line Summary

Ubuntu Server is a lightweight, secure, and enterprise-ready Linux operating system designed to host network services and applications. Its stability, long-term support, and compatibility make it an excellent choice for cybersecurity home labs and production environments.
